

★ WEEK 3 · LECTURE

Labeling & Meta-Labeling

Teaching the model WHAT to predict — with a clever answer key

Expanded edition · 38 slides · with three hands-on labs

A model is only as good as the question we ask it. This week we learn to write the 'answer key' the right way — using a race with three finish lines.

Where We Are Going Today

Week 1 = what ML is · Week 2 = clean data (bars, CUSUM) · Week 3 = the answer key

1

What is a 'label', and why does it matter?

It's the answer key. Get it wrong and even a perfect model learns the wrong thing.

2

How do pros label a trade? (Triple-Barrier)

A race with three finish lines: profit, loss, and a time limit.

3

How do we decide how BIG to bet? (Meta-Labeling)

Split the work between two specialist models — a scout and a manager.

By the end you will label real trades and reshape the answer key in your browser.

1

PART 1 • THE ANSWER KEY

What Is a Label?

The answer key we hand the model

Remember the Flashcards?

Supervised learning needs the back of every card filled in

Back in Week 1: supervised learning = flashcards with a question (X) and an answer (y). The 'label' is that answer on the back of the card. No labels → nothing to learn from.

The catch in finance

- For a photo of a cat, the label is obvious: 'cat'.
- For a stock, what is the 'right answer'?
- 'Did it go up?' sounds simple — but it's a trap.

Today's mission

- Learn to write labels that match how trading really works.
- A good label = a model that learns something useful.
- A lazy label = a model that learns nonsense.

"Will the Price Be Higher Tomorrow?"

López de Prado: that famous question is the wrong one to ask

"The most common way beginners label data is also the most misleading.

Time is not how traders win or lose."

The trap

- Label = 'is the price up after exactly 5 days?'
- But on day 2 you might be down 20% and forced to sell.
- The label says 'up' — yet you already lost. It lies about your real experience.

The fix (this week)

- Label by what actually ends a trade: a profit target, a stop-loss, or a deadline.
- This is the Triple-Barrier method.
- It bakes risk into the answer key itself.

Lazy Question vs. Smart Question

A good label mirrors how a real trader actually enters and exits a position.

Labeling = Writing the Answer Key First

Lazy question (what beginners ask)

"Will the price be higher in exactly 5 days?"

Ignores the wild ride in between.

You might be down 20% on day 2 and stopped out — but the label still says "up". Misleading!

No risk, no path, no realism.

Smart question (what pros ask)

"If I buy with a profit target and a stop-loss, which hits first?"

Mirrors how a real trader exits.

Captures risk AND reward AND the path the price actually took.

This is the Triple-Barrier method — our answer key for the model.

Three Ways 'Up in 5 Days?' Lies

Each blind spot has a fix — together the fixes ARE the Triple-Barrier

Blind spot	Why it hurts	The fix (coming up)
Blind to the path	A trade stopped out on day 2 still gets labeled a 'win'.	Watch every day until a finish line is touched.
Blind to risk	No stop-loss exists in that world — losses are invisible.	Put the stop-loss INSIDE the label itself.
One clock for all	Why exactly 5 days? Week 2 déjà vu: time is a bad ruler.	Profit / loss exits first; time is only the backstop.

Sound familiar? The Week 2 lesson — 'the clock is a bad ruler' — now applied to the answer key.

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PART 2 · TRIPLE-BARRIER

The Triple-Barrier Method

A race with three finish lines

Three Finish Lines for Every Trade

After you buy, three lines race to be touched first — the first touch becomes the label: +1, -1, or 0.

The Triple-Barrier Method — Three Finish Lines



Whichever line the price touches FIRST decides the label. Here it hit profit-take, so $y = +1$.

Meet the Three Barriers

Each is just a line on the price chart — first touch wins

Top line — Profit-take

label = +1

Set above your buy price. If price rises and touches it first, the trade was a WIN.

like a 'take profit' order

Bottom line — Stop-loss

label = -1

Set below your buy price. If price falls and touches it first, the trade was a LOSS.

like a 'stop loss' that limits damage

Right line — Time limit

label = 0

A deadline (e.g. 10 days). If neither price line is hit by then, the trade just expired.

stops us waiting forever

First touch wins. That single rule turns a messy price path into one clean label.

The Same Rule, as a Formula

It just lists the three possible answers

$$y = \begin{cases} +1 & \text{hit profit-take first} \\ -1 & \text{hit stop-loss first} \\ 0 & \text{time runs out} \end{cases}$$

How to read it

- y is the label — the answer the model will try to predict.
- It can only be one of three values: +1, -1, or 0.
- The curly brace just means 'pick the case that happened first'.

Why three values, not two

- Up/down alone ignores 'nothing really happened'.
- The 0 case protects us from forcing a bet on noise.
- Three honest outcomes beat two forced ones.

What Each Label Teaches the Model

Three answers, three different lessons — and the 0 is not garbage

Label	When it happens	What the model learns from it
+1	Profit-take touched first	This setup tends to pay before it hurts.
-1	Stop-loss touched first	This setup tends to hurt first — avoid or fade it.
0	Time limit expires untouched	This setup goes nowhere — dead money, skip it.

The 0 class is a gift: 'no trade' is a real decision that 'up in 5 days?' can never express.

How Wide Should the Lines Be?

The widths are knobs you choose — and they change everything

Set the profit and loss lines using recent volatility — how much the stock usually moves. A calm stock gets narrow lines; a wild stock gets wide ones. One size does NOT fit all.

Symmetric (1:1)

- Profit line and stop-loss equally far away.
- Treats upside and downside the same.
- A neutral starting point.

Asymmetric (e.g. 2:1)

- Profit line twice as far as the stop.
- Says: 'cut losses fast, let winners run'.
- Encodes a real trading philosophy into the label.

Why the Lines Must Fit the Stock

A fixed $\pm 2\%$ wall is two different walls on two different stocks

Fixed $\pm 2\%$ on everything

- Sleepy utility stock (moves 0.5%/day): $\pm 2\%$ is a fortress — almost every label becomes 0.
- Wild small-cap (moves 4%/day): $\pm 2\%$ is tissue paper — pure noise triggers ± 1 labels.
- The label's meaning silently changes from stock to stock.

Volatility-scaled lines

- Measure each stock's usual daily move first.
- Set the lines as a multiple of THAT — e.g. $2\times$ its typical move.
- 'A real move' now means the same thing on every chart.

Same idea as Week 2's CUSUM threshold in σ units — one rule, every stock.

Three Presets for the Knobs (pt, sl)

The barrier shape encodes your trading style before any model runs — Lab 2 sweeps exactly these

(+2%, -2%) — neutral

symmetric 1:1

No opinion on direction. Labels measure pure follow-through.
Start here in Lab 1.

research baseline

(+3%, -1%) — trend style

let winners run

Wide target, tight stop. More -1 labels — but the +1s that survive are real moves.

momentum philosophy

(+1%, -3%) — quick-profit style

take gains fast

Tight target, wide stop. Many quick +1s, occasional deep -1 — a mean-reversion flavor.

market-making flavor

The label already contains a strategy assumption — write it down, don't inherit it by accident.

One Trade by Hand

Buy at 100. Lines: profit-take 104, stop-loss 96, time limit 6 days (full version: this week's problem set)

Day	Price	vs profit-take 104	vs stop-loss 96	Verdict
1	101.0	below	above	race continues
2	102.0	below	above	race continues
3	103.0	below	above	race continues
4	104.5	TOUCHED ✓	above	label = +1 · race over — days 5–6 never checked

First touch ends the race — the remaining days are irrelevant, exactly like a real exit.

A Year Through the Three Finish Lines

Ballpark shares you will verify in Lab 1 — ($\pm 2\%$, 10-day limit) on a large-cap stock

~40%

+1 labels — the profit-take wins the race

~35%

-1 labels — the stop-loss wins

~25%

0 labels — the clock wins: nothing happened

90% 0s?

Your lines are too wide for this stock — retune before modeling

The mix is a health check: extreme shares mean the knobs, not the market, are speaking.

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PART 3 · META-LABELING

Meta-Labeling

Two specialists: direction, then size

One Model Trying to Do Two Jobs

Predicting direction AND deciding how much to bet is too much for one model

Imagine one person being both the talent scout AND the team manager. Better to split: one finds opportunities, the other decides which to actually back — and how hard.

The two jobs

- Job 1 — direction: will it go up or down?
- Job 2 — confidence: how sure are we, so how big should the bet be?
- These need different skills.

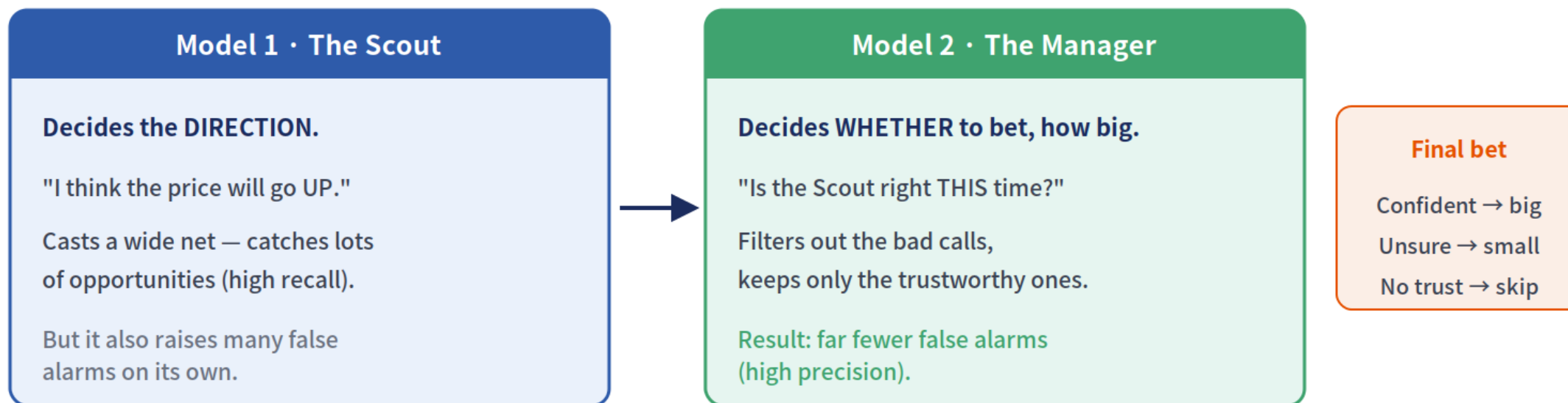
Meta-Labeling splits them

- Model 1 predicts the direction (the 'scout').
- Model 2 predicts whether Model 1 is right this time (the 'manager').
- Each does one job well.

Meta-Labeling — a Scout and a Manager

Model 1 casts a wide net (recall); Model 2 decides what to keep (precision).

Meta-Labeling — A Team of Two Specialists



Split the job: one model picks the direction, the other decides how much to trust it.

Precision vs. Recall — the Fishing Net

Model 1 chases recall; Model 2 adds precision. Together they beat either one alone.

Precision vs. Recall — Two Ways to Be Right

Think of a fishing net

Recall = "Did I catch all the fish?"

Of every real fish in the sea,
how many did my net grab?

Precision = "Is my catch clean?"

Of everything in my net, how
much is real fish, not junk?

A huge net catches all fish (recall)
but also rocks and boots (low precision).

The 2x2 outcome grid

True Positive (correct hit)	False Positive (false alarm)
False Negative (missed it)	True Negative (correct pass)

Columns = what the model SAID
Rows = what was actually TRUE

Scoring the Balance: the F1 Score

One number that rewards getting BOTH precision and recall right

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}$$

Reading it

- Precision: of the trades we took, how many were good?
- Recall: of all good trades out there, how many did we catch?

Why we like F1 here

- Can't cheat it by only being safe (high P, low R) or betting on everything (high R, low P).
- It is high only when BOTH are high — an honest balance.

Four Signals Through the Manager

The scout proposes, the manager disposes — probability becomes position size

Event	Scout says	Manager $P(\text{scout right})$	Decision
Mon	LONG	0.72	full-size long
Tue	SHORT	0.55	half-size short
Wed	LONG	0.31	SKIP — signal fired, wallet stays shut
Thu	SHORT	0.64	two-thirds short

The scout never changes its mind — the manager just decides how much that mind is worth.

When Meta-Labeling Shines

Four setups where the scout-and-manager split earns its keep

Setup	Why the split helps
A human PM's discretionary calls	Keep the human's edge; let ML size the conviction objectively.
A simple white-box rule (e.g. momentum)	The rule stays auditable; ML only filters its weak moments.
A signal with good recall, poor precision	The manager removes false positives — the exact failure mode.
Regulated settings needing explainability	Direction logic stays transparent; only the sizing is learned.

Notice the theme: meta-labeling ADDS ML on top of judgment instead of replacing it.

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PART 4 · APPLICATION

Where This Shows Up

Fraud detection uses this exact trick

Meta-Labeling in Fraud Detection

The same scout-and-manager idea, in a bank

Fraud is rare (~1 in 1,000). Banks must catch it WITHOUT blocking honest customers. So they use exactly the two-model split we just learned.

Model 1 — the wide net

- A simple, fast rule flags anything suspicious.
- Goal: miss as little real fraud as possible (high recall).
- Downside: it also flags many honest payments.

Model 2 — the filter

- A smarter model reviews each flag.
- Goal: block real fraud, wave through the honest ones (high precision).
- Result: fewer angry customers, less missed fraud.

Label It! — A 60-Second Drill

Cover the right column, answer out loud, then check yourself

Scenario (buy at 100, lines 104 / 96, limit 6 days)	Label / concept?
Path: 102, 103, 104.2 on day 3	+1 — profit-take touched first
Path: 99, 97, 95.5 on day 3	-1 — stop-loss touched first
Price wanders 99–103 for six days	0 — the time limit wins the race
Same path, knobs (+1%, -3%) instead of ($\pm 2\%$)	A different label — the knobs ARE the question
Manager outputs $P(\text{scout right}) = 0.3 \rightarrow \text{skip}$	Meta-labeling — precision over pride

If you got 4 of 5, the vocabulary of Week 3 is yours.

The Assembly Line, Three Stations Long

Week 2 built the rows; Week 3 wrote the answers; Week 4 finally gets to model

Bars (rows) → Events (which rows) → Labels (the answer key — today) → Models (Week 4)

What is now locked in

- Activity-sampled bars and CUSUM events (Week 2).
- Risk-aware Triple-Barrier labels (today).
- A scout-and-manager plan for direction vs. size (today).

What Week 4 will assume

- A clean (X, y) table — features per event, label from the race.
- The 0 class as a legitimate 'no trade' answer.
- Only now do we argue about models — question first, always.

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PART 5 · HANDS ON

Your Turn — Hands On

Label real trades and reshape the answer key in your browser

Getting This Week's Data

Companion notebook — Week 3 loader is in Section 3(b)

From the companion notebook

- Open FML_Data_Setup_Guide.ipynb in Colab.
- Run Setup (§ 0), then the § 3 price loader.
- Offline backup (§ 5) works if internet is blocked.

What you need

- A price series (Close) — that's it.
- We pick a profit %, a stop %, and a day limit.
- Then walk forward to see which line is hit first.

[CODE] FML_Data_Setup_Guide.ipynb · § 3 price loader

```
import yfinance as yf, numpy as np
px = yf.download('AAPL', period='2y', auto_adjust=True,
                 multi_level_index=False, progress=False)['Close']
# we will attach Triple-Barrier labels to px in Lab 1
```

Hands-On Lab 1 — Label a Trade

Turn a price series into +1 / -1 / 0 labels

[CODE] colab_lab1.py · data from setup guide § 3

```
import numpy as np
px = px.dropna(); pt, sl, limit = 0.02, -0.02, 10
labels = []
for i in range(len(px) - limit):
    p0 = px.iloc[i]
    window = px.iloc[i+1 : i+1+limit] / p0 - 1
    y = 0
    for r in window:
        if r >= pt: y = 1; break
        if r <= sl: y = -1; break
    labels.append(y)

import pandas as pd
print(pd.Series(labels).value_counts())
```

Your task (step by step)

- Load prices (setup guide § 3).
- Choose $pt = +2\%$, $sl = -2\%$, $limit = 10$ days.
- For each day, look forward up to 10 days.
- See which barrier the price hits first.
- Record +1, -1, or 0.
- Q: how many of each label did you get?

Hands-On Lab 2 — Change the Lines

Watch the labels shift: barrier widths reshape the answer key

[CODE] colab_lab2.py · data from setup guide § 3

```
import numpy as np, pandas as pd
px = px.dropna(); limit = 10

def label_all(pt, sl):
    out = []
    for i in range(len(px) - limit):
        w = px.iloc[i+1:i+1+limit] / px.iloc[i] - 1
        y = 0
        for r in w:
            if r >= pt: y = 1; break
            if r <= sl: y = -1; break
        out.append(y)
    return pd.Series(out).value_counts().to_dict()

for pt, sl in [(0.02, -0.02), (0.03, -0.01), (0.01, -0.03)]:
    print(pt, sl, '->', label_all(pt, sl))
```

Your task (step by step)

- Reuse your labeling code from Lab 1.
- Try (2%, -2%), (3%, -1%), (1%, -3%).
- Count the labels for each setting.
- Q: how does 'let winners run, cut losses' (3%, -1%) change the mix?

Hands-On Lab 3 — Claude as Your Coding Copilot

AI is allowed in this course — it explains and extends; you understand and verify

[INPUT] Prompt · Claude (paste after your Lab 1 + 2 code)

You are my machine-learning tutor.

- 1) Explain my labeling loop above, one plain-English sentence per line.
- 2) Build a SCOUT: side = sign of the 20-day momentum on each labeled day.
- 3) Meta-label each day: 1 if the scout's side matched the Triple-Barrier label, else 0 (drop the 0-label days).
- 4) Fit sklearn LogisticRegression on features [abs(momentum), 20-day volatility] to predict the meta-label (70/30 time-ordered split, no shuffling!).
- 5) Print precision, recall and F1 of: all scout bets vs. bets the manager approves ($P > 0.5$), and explain the gap in 3 sentences using scout-and-manager language.

House rules for AI use

- Allowed: explain, debug, extend code you wrote.
- Required: run it, read it, defend every line.
- Forbidden: submitting code you cannot explain.
- Tip: ask 'why', not just 'do'.

Comprehension Check

Try each question first — the answer is revealed below

Q1 In Triple-Barrier, what decides the label?

- A** The price after a fixed number of days
- B** **Whichever barrier the price touches first** ✓
- C** The average price over the window
- D** The trading volume

Answer: First touch wins: profit-take $\rightarrow +1$, stop-loss $\rightarrow -1$, time limit $\rightarrow 0$.

Q2 In Meta-Labeling, the second model mainly improves:

- A** Recall (catching everything)
- B** **Precision (avoiding false alarms)** ✓
- C** Download speed
- D** The number of features

Answer: Model 1 casts a wide net (recall); Model 2 filters out bad calls, raising precision.

Comprehension Check — Round Two

Two more — these cover the new slides

Q3 You switch knobs from ($\pm 2\%$) to ($+3\%$, -1%). The label mix will show:

- A** More 0s — nothing changes
- B** More -1 s: the tight stop is hit often ✓
- C** Only $+1$ s
- D** The same mix — knobs don't matter

Answer: A stop at -1% is touched easily; the surviving $+1$ s need a full $+3\%$ run. The mix shifts toward -1 .

Q4 A strategy has precision 90%, recall 10%. Its F1 is:

- A** 90% — precision is what counts
- B** 50% — just the average
- C** Low ($\sim 18\%$) — F1 punishes imbalance ✓
- D** Undefined

Answer: $F1 = 2PR/(P+R) = 2(0.9)(0.1)/1.0 = 18\%$. Being 'safe but blind' cannot cheat the F1 score.

Two Big Ideas, One Picture Each

A great model starts with a great QUESTION — label well, and you're halfway there

1

Triple-Barrier = a race with three finish lines

Profit-take (+1), stop-loss (−1), time limit (0). First touch wins. Risk is baked into the label.

2

Meta-Labeling = a scout and a manager

Model 1 picks the direction (recall); Model 2 decides whether to bet and how big (precision).

Keep two pictures in mind — three finish lines, and a scout-plus-manager team.

Week 3 — Three Things to Remember

Before next class: finish Labs 1–3 and note how the label mix changed with the barriers

1

The label is the answer key

'Will it be up in 5 days?' is a trap. Label by how a trade really ends — with risk built in.

2

Triple-Barrier: profit, loss, or time

Three finish lines. First touch decides +1, −1, or 0. Set the widths from each stock's volatility.

3

Meta-Labeling splits direction from size

One model finds opportunities (recall); a second decides how much to trust each one (precision).

Next week — Linear Models & Regularization: taming the 'factor zoo' with Lasso.

Week 3 on One Page

Photograph this slide — it is the whole lecture

Concept	One line to remember
Lazy label	'up after h days?' — blind to path, risk, and the stock's temperament.
Triple-Barrier	profit-take / stop-loss / time limit; first touch $\rightarrow +1 / -1 / 0$.
The knobs (pt, sl, T)	scale with volatility; the ratio encodes a trading philosophy.
Meta-label	1 if the scout's side was right, else 0 — the manager's training data.
Precision · Recall · F1	took-and-right · caught-of-all-good · the honest blend of both.

Everything downstream — models, backtests, sizing — stands on these five lines.